

Lecture 1 — Inductive Definitions (1)

COSE212: Programming Languages

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Inductive Definitions

Inductive definition (induction) is widely used in the study of programming languages and computer science in general: e.g.,

- The syntax and semantics of programming languages
- Data structures (e.g., lists, trees, graphs)

Induction is a technique for formally defining a set:

- The set is defined in terms of itself.
- The only way of defining an infinite set by a finite means.

Three styles to inductive definition:

- Top-down
- Bottom-up
- Rules of inference

Example (Top-Down)

Let us define a certain subset $\mathbf{S}$ of natural numbers ($\mathbb{N}$) as follows:

Definition ($\mathbf{S}$)

A natural number $\mathbf{n}$ is in $\mathbf{S}$ if and only if

1. $\mathbf{n} = 0$, or
2. $\mathbf{n} - 3 \in \mathbf{S}$.

The definition is *inductive*, because the set is defined in terms of itself. What is the set $\mathbf{S}$?

Example (Continued)

Let us see what natural numbers are in $\mathcal{S}$.

- 0 is in $\mathcal{S}$ because of the first condition of the definition.
- 3 is in $\mathcal{S}$ because $3 - 3 = 0$ and 0 is in $\mathcal{S}$.
- 6 is in $\mathcal{S}$ because $6 - 3 = 3$ and 3 is in $\mathcal{S}$.
- ...

We can conjecture that $\{0, 3, 6, 9, \dots\} \subseteq \mathcal{S}$.

Proof by mathematical induction .

We show that $3k \in \mathcal{S}$ for all $k \in \mathbb{N}$.

1. Base case: $3k \in \mathcal{S}$ when $k = 0$.
2. Inductive case: Assume $3k \in \mathcal{S}$ (Induction Hypothesis, I.H.).
Then show $3 \cdot (k + 1) \in \mathcal{S}$, which holds because $3 \cdot (k + 1) - 3 = 3k \in \mathcal{S}$ by the induction hypothesis.



Example (Continued)

What about other numbers? Does $\mathbf{S}$ contain only the multiples of 3?

- For instance, $1 \in \mathbf{S}$? No. Because the first condition is not true, the second condition must be true for 1 to be in $\mathbf{S}$. However, it is not true because $1 - 3 = -2$ is not a natural number. Similarly, we can show that $2 \notin \mathbf{S}$.
- What about 4? Because $4 - 3 = 1 \notin \mathbf{S}$, $4 \notin \mathbf{S}$.

By similar reasoning, we can conjecture that if n is not a multiple of 3 then n is not in $\mathbf{S}$. In other words, $\mathbf{S}$ contains multiples of 3 only: i.e.,

$$\{0, 3, 6, 9, \dots\} \supseteq \mathbf{S}.$$

Proof by contradiction.

Let $n = 3k + q$ ($q = 1$ or 2) and assume $n \in \mathbf{S}$. By the definition of $\mathbf{S}$, $n - 3$, $n - 6$, $\dots$, $n - 3k \in \mathbf{S}$. Thus, $\mathbf{S}$ must include 1 or 2, a contradiction. □

A Bottom-up Definition

An alternative inductive definition of S :

Definition (S)

S is the *smallest* set such that $S \subseteq \mathbb{N}$ and S satisfies the following two conditions:

1. $0 \in S$, and
2. if $n \in S$, then $n + 3 \in S$.

- The two conditions imply $\{0, 3, 6, 9, \dots\} \subseteq S$.
- The two conditions do not imply $\{0, 3, 6, 9, \dots\} \supseteq S$. E.g.,
 - $\mathbb{N}$ satisfies the conditions: $0 \in \mathbb{N}$ and if $n \in \mathbb{N}$ then $n + 3 \in \mathbb{N}$.
 - $\{0, 3, 6, 9, \dots\} \cup \{1, 4, 7, 10, \dots\}$ satisfies the conditions.
- This is why the definition requires S to be the **smallest** such a set.
- The smallest set that satisfies the two conditions is unique:

$$S = \{0, 3, 6, 9, \dots\}.$$

Rules of Inference

The third way is to define the set with inference rules. An inference rule is of the form:

$$\frac{A}{B}$$

- **A**: hypothesis (antecedent)
- **B**: conclusion (consequent)
- “if **A** is true then **B** is also true”.
- $\overline{B}$: axiom (inference rule without hypothesis)

The hypothesis may contain multiple statements:

$$\frac{A \quad B}{C}$$

“If both **A** and **B** are true then so is **C**”.

Rules of Inferences

The set $\mathcal{S}$ is defined as inference rules as follows:

Definition ($\mathcal{S}$)

$$\overline{0 \in \mathcal{S}} \quad \frac{n \in \mathcal{S}}{(n+3) \in \mathcal{S}}$$

Interpret the rules as follows:

“A natural number n is in $\mathcal{S}$ iff $n \in \mathcal{S}$ can be derived from the axiom by applying the inference rules finitely many times”

For example, $3 \in \mathcal{S}$ because we can find a “proof/derivation tree”:

$$\frac{\overline{0 \in \mathcal{S}}}{3 \in \mathcal{S}} \begin{array}{l} \text{the axiom} \\ \text{the second rule} \end{array}$$

but $1, 2, 4, \dots \notin \mathcal{S}$ because we cannot find proofs. Note that this interpretation enforces that $\mathcal{S}$ is the smallest set closed under the inference rules.

Exercises

1. What set is defined by the following inductive rules?

$$\frac{}{3} \quad \frac{x \quad y}{x + y}$$

2. What set is defined by the following inductive rules?

$$\frac{}{()} \quad \frac{x}{(x)} \quad \frac{x \quad y}{xy}$$

3. Define the following set as rules of inference:

$$S = \{a, b, aa, ab, ba, bb, aaa, aab, aba, abb, baa, bab, bba, bbb, \dots\}$$

4. Define the following set as rules of inference:

$$S = \{x^n y^{n+1} \mid n \in \mathbb{N}\}$$

Summary

In inductive definitions, a set is defined in terms of itself. Three styles:

- Top-down
- Bottom-up
- Rules of inference

In PL, we mainly use the rules-of-inference method.